University of Colombo School of Computing

SCS 1304 - Problem Solving Strategies and Computation Approaches

Lab Sheet 06

A. Consider the following code snippet and identify the basic operation first. Write the time complexity (Big O) of the code. Show your calculations.

1.
```
FOR i FROM 0 TO n-1
  FOR j FROM 0 TO i
    PRINT i, j
```

2.
```
FOR i FROM 0 TO n-1
  FOR j FROM 0 TO m-1
    FOR k FROM 0 TO p-1
      PRINT i, j, k
```

3.
```
FOR i FROM 0 TO n-1
  FOR j FROM 0 TO i
    IF (i % 2 == 0)
      PRINT i, j
    ELSE
      FOR k FROM 0 TO j
        PRINT i, j, k
```

4.
```
i = 1
WHILE i < n
  FOR j FROM 0 TO i
    PRINT i, j
  i = i * 2
```

5.
```
FOR i FROM 1 TO n
  FOR j FROM 1 TO i * i
    IF (j % i == 0)
      FOR k FROM 0 TO j
        PRINT i, j, k
```

6.
```
FOR i FROM 0 TO n-1
  k = i
  WHILE k < n
    PRINT i, k
    k = k + i + 1
```

B. Define basic operations for your algorithms in the given below Exercises 1–3, and study the performance of these algorithms. Determine the worst-case time complexity.

   1. Write a pseudocode algorithm that finds the largest number in a list (an array) of n numbers.
   2. Write a pseudocode algorithm that finds the m smallest numbers in a list of n numbers.
   3. Write a pseudocode algorithm that prints out all the subsets of three elements of a set of n elements. The elements of this set are stored in a list that is the input to the algorithm.

C. Suppose you have a computer that requires **1 minute** to solve problem instances of size **n = 1,000**. Suppose you buy a new computer that runs **1,000 times faster** than the old one. What instance sizes can be run in **1 minute**, assuming the following time complexities T(n)T(n)T(n) for our algorithm?

   (a) $T(n)=n$

   (b) $T(n)=n^3$

   (c) $T(n)=10^n$

D. The function : $\mathbf{f(n)=3n^2+10n \log n+1000n+4 \log n+9999}$ belongs to which of the following complexity categories?

   (a) $\theta(\log n)$ (b) $\theta(n^2 \log n)$ (c) $\theta(n)$ (d) $\theta(n \log n)$ (e) $\theta(n^2)$

   (f) None of these

E. Consider the following algorithm:

```
int add_them (int n, int A[])
    {
      index i, j, k;
      j = 0;
      for (i = 1; i <= n; i++)
        j = j + A[i];

      k = 1;
      for (i = 1; i <= n; i++)
        k = k + k;

            return j + k;
    }
```

Answer the following questions:

1. If n=5 and the array A contains the values **2, 5, 3, 7, and 8**, what is the output of the algorithm?
2. What is the time complexity T(n) of the algorithm? Justify your answer by identifying the number of times each loop executes.
3. Try to improve the efficiency of the algorithm. Write an improved version and determine its time complexity.